

Zibra AI: Revolutionizing VFX Production with Real-Time Volumetric Effects

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Figure 1: Volumetric Smoke & Fire rendered in real-time in UE

ABSTRACT

Zibra AI introduces a unified approach, integrating both the creation and rendering of VFX directly within game engines, aiming to streamline and enhance the production process. The new workflow is empowered by two major innovations: ZibraVDB — an OpenVDB compression tool that enables real-time rendering of volumetric VFX, and Zibra Effects — a highly performant real-time VFX simulation technology. Both tools support in-engine workflows for Unreal Engine, Unity, or any other custom engine. ZibraVDB achieves up to a 150x compression rate while maintaining the high visual fidelity required for advanced production workflows such as virtual production and AAA game development. Combined with a highly

efficient custom render pass, it achieves significant rendering performance. Zibra Effects enables the physics-based VFX simulation of liquids, smoke, and fire directly inside a game engine. It leverages proprietary AI-based SDF compression technology that reduces collider memory footprint by 40x on average. In combination with ZibraVDB, it covers the VFX creation workflow end-to-end: users can choose between real-time interactive effects or cache them into ZibraVDB format for efficiency

CCS CONCEPTS

• Real-time systems; • Computer graphics; • Machine learning;

KEYWORDS

VDB compression, SDF compression, Volumetric visual effects, Real-time visual effects simulation

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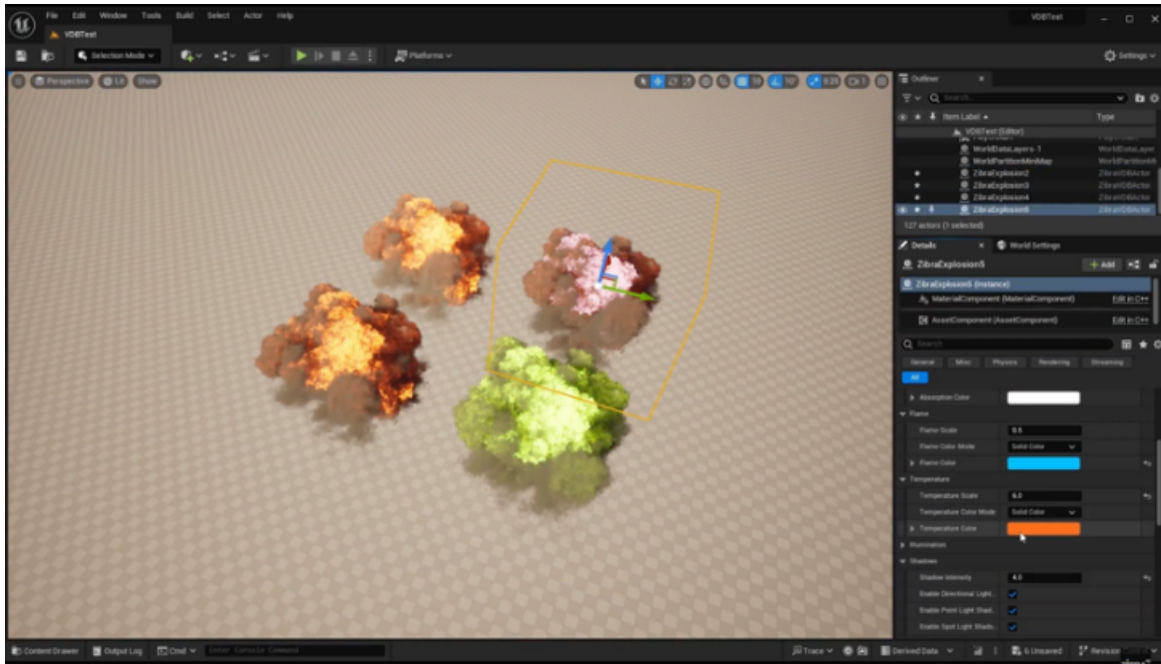


Figure 2: Replaying VDBs in UE using Zibra VDB

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1 INTRODUCTION

Zibra AI’s tools enable a seamless and efficient workflow for VFX creation inside the game engine. The key contributions of this paper are as follows:

- Introducing the technology stack that includes the ZibraVDB solution for compression and the Zibra Effects solution for simulation.
- Presenting an integrated workflow of real-time and cached VFX.
- Demonstrating the seamless integration into real-time engines.
- Showcasing the benefits of the new workflow.

2 VDB COMPRESSION IN ACTION

ZibraVDB demonstrates its powerful compression capabilities with up to a 150x compression rate, offering an option to choose between quality levels and compression rates, allowing customers to adjust for specific use cases. Substantial performance enhancements compared to Unreal Sparse Volumetric Textures (SVT) show significant FPS improvements and scalability, emphasizing the technology’s ability to handle extensive VDB sequences efficiently.

3 ZIBRA AI REAL-TIME SIMULATION CAPABILITIES

Real-time, highly performant simulations of liquids, smoke, and fire were created with an individual approach when developing key algorithms, allowing for the real-time simulation of millions of particles, covering immersive experiences when interaction is required. Empowered with another proprietary AI-based SDF compression algorithm that reduces memory footprints and creates efficient colliders, enhancing overall performance and interactivity.

4 ZIBRA AI INTEGRATED WORKFLOW

The versatility of the Zibra AI toolkit offers users the choice between baking effects with Zibra VDBx technology for efficiency or leveraging real-time interactive effects for a more immersive experience. The capability to replay cached VDB sequences within both Unity and Unreal engines highlights Zibra AI’s adaptability, cross-platform compatibility, and first steps into VFX interoperability.